

Chapter 10: Symmetrical Designs

Worksheet 10: Symmetrical Designs

Learning Objective: Identify lines of symmetry in shapes and designs; complete symmetric patterns; relate symmetry to art and festival designs such as rangoli.

Worked Example:

How many lines of symmetry does a square have?

Solution: A square has 4 lines of symmetry: 2 diagonals, 1 horizontal and 1 vertical.

Questions (1–20)

1. How many lines of symmetry does a circle have?
2. Does the letter “A” have a line of symmetry? If yes, where?
3. How many lines of symmetry does an equilateral triangle have?
4. Is the letter “F” symmetrical?
5. Name two capital letters that have a vertical line of symmetry.
6. A rectangle (not a square) has how many lines of symmetry?

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7. Half of a butterfly's wing pattern is drawn. If the design is symmetric, describe what the other half should look like.

 8. Does a regular pentagon have line symmetry? If so, how many lines?

 9. Which has more lines of symmetry: a square or a rectangle?

 10. Name one design from your home (rangoli, mehendi, or a window grill) that shows symmetry.

 11. A rangoli design has 6 identical petals arranged evenly around a centre point, like a flower. How many lines of symmetry would this design have?

 12. One half of a kite shape is drawn, symmetric about a vertical line. Describe how to complete the other half.

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13. A regular hexagon has how many lines of symmetry?
14. Which has more lines of symmetry: a regular pentagon or a regular hexagon?
15. Is the number “8” symmetrical when written normally? Explain.
16. A student designs a rangoli with a vertical line of symmetry. She has drawn 3 red dots and 2 blue dots on the left half. What must she draw on the right half to complete the symmetric design?
17. A wallpaper border repeats a symmetric flower motif every 15 cm along a wall that is 3 m long. How many times will the motif repeat?
18. Two students fold paper in half and cut a shape along the fold. What kind of symmetry will the unfolded design always have, and why?

19. Find a symmetric design in your home or neighbourhood (a window, rangoli, cloth, or gate). Describe it and mark where the line of symmetry is.
20. During a festival, families often draw rangoli at the entrance. Describe in words a simple rangoli pattern that has at least one line of symmetry.

Reflection Question: Why do you think many traditional designs (rangoli, mehendi, embroidery) are made symmetric? Write your own idea.

Real-Life Application Question: Fold a piece of paper in half, cut a simple shape along the fold, and open it. Describe the symmetric design you created.